Project 2:

Data Analysis Plan

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Data Management and Cleaning

Data will be screened for outliers, missingness, correlations, and distributional assumptions using graphical summaries and descriptive statistics, with transformations applied as needed. Participants excluded due to missing data will be compared with the full cohort to evaluate potential biases.

General Analytical Framework

Linear regression models will estimate associations between inflammatory markers and one-year changes in each outcome, allowing adjustment for covariates and testing of interaction effects.

To address multiplicity, analyses will be organized into prespecified hypothesis families corresponding to H1a, H1b, and Aim 2. Because the cognitive measures are expected to be intercorrelated, they will be treated as tests of the same hypothesis rather than independent tests, and a single cognitive outcome model will be estimated. Cortical thickness and cognitive outcomes, however, showed weak preliminary correlations and are therefore will be treated as independent.

Because preliminary results also showed strong correlation between inflammatory markers(IL-6 and MCP-1), the family-wise type I error rate will be controlled at two-sided $\alpha = 0.0125$ ($\alpha = 0.05 / 4$ independent tests). If correlations among tests are weaker than anticipated, alpha will be further adjusted to reflect more independent tests.

Details of the Analyses

Primary outcomes include one-year changes in cognitive performance and AD-signature cortical thickness (five outcomes total). Primary exposures include circulating inflammatory markers measured at baseline and one-year follow-up. All models will adjust for age, sex, APOE genotype, baseline Clinical Dementia Rating (CDR), and baseline outcome value, consistent with variables collected in the parent RMADC studies.

For Hypothesis H1a, models will evaluate associations between baseline inflammatory markers and one-year outcome changes. For Hypothesis H1b, models will evaluate associations between changes in inflammatory markers and outcome changes.

For Aim 2, models will include amyloid deposition and an inflammation x amyloid interaction term. Amyloid will be dichotomized using an SUVR threshold of 1.10 (low vs high).

Sensitivity analysis will be performed to test whether covariates are potential confounders, mediators, or precision variables for each outcome.

Summary of Analyses

Aim	Hypothesis	Primary Outcome	Primary Exposure	Covariates
1	H1a	1-yr Δ in CVLT, Benson Figure, MST, and Store Recall (continuous cognitive measures); 1-yr Δ in AD-signature cortical thickness (continuous)	Baseline inflammatory markers (continuous cytokines: IL-6, TNF-α, continuous chemokines:MCP-1, Eotaxin-1)	Baseline age, sex, & baseline outcome
1	H1b	Same outcomes	Change in inflammatory markers (baseline → year-1)	Baseline age, sex, baseline outcome, and baseline inflammatory marker level
2	H2	Same outcomes	baseline Inflammation × baseline amyloid interaction	Same as H1b

Power Analysis

Power calculations assumed a final analytic sample size of N = 175. For Aim 1, power was estimated using correlation-based methods (powertools::corr.1samp in R) across a range of significance thresholds reflecting uncertainty in the effective number of independent tests (α = 0.05/4, 0.05/10, 0.05/20). Minimum detectable correlations required to achieve 80% power were similar across thresholds (|r| ≈ 0.26–0.29; **Table 1**), indicating that the choice of alpha had little bearing on conclusions about adequacy of power.

Aim 1 Power at N = 175

Power (%) across assumed correlations and significance thresholds

	Power (%)		
Assumed |r|	**α = 0.05/4**	**α = 0.05/10**	**α = 0.05/20**
0.50	100.0	100.0	100.0
0.45	100.0	100.0	100.0
0.40	99.9	99.7	99.5
0.35	99.0	97.7	96.3
0.30	94.2	89.7	85.3
0.25	80.6	71.0	63.2
0.20	56.7	44.4	36.1
Min |r| for 80% power	**25.0**	**28.0**	**29.0**

Minimum detectable correlation corresponds to the smallest absolute correlation achieving at least 80% power.

Table 1. Aim 1 Power Calculations at N = 175. *Estimated statistical power (%) to detect correlations between inflammatory markers and outcome changes across a range of assumed effect sizes and significance thresholds reflecting different multiplicity assumptions (α = 0.05/4, 0.05/10, 0.05/20).*

Accordingly, Aim 2 power analyses were conducted at α = 0.01. Power was estimated using two-sample correlation methods (powertools::corr.2samp in R), comparing associations between inflammation and outcomes across amyloid-negative and amyloid-positive groups. Power was evaluated across plausible ranges of correlations and group size

distributions. Results (

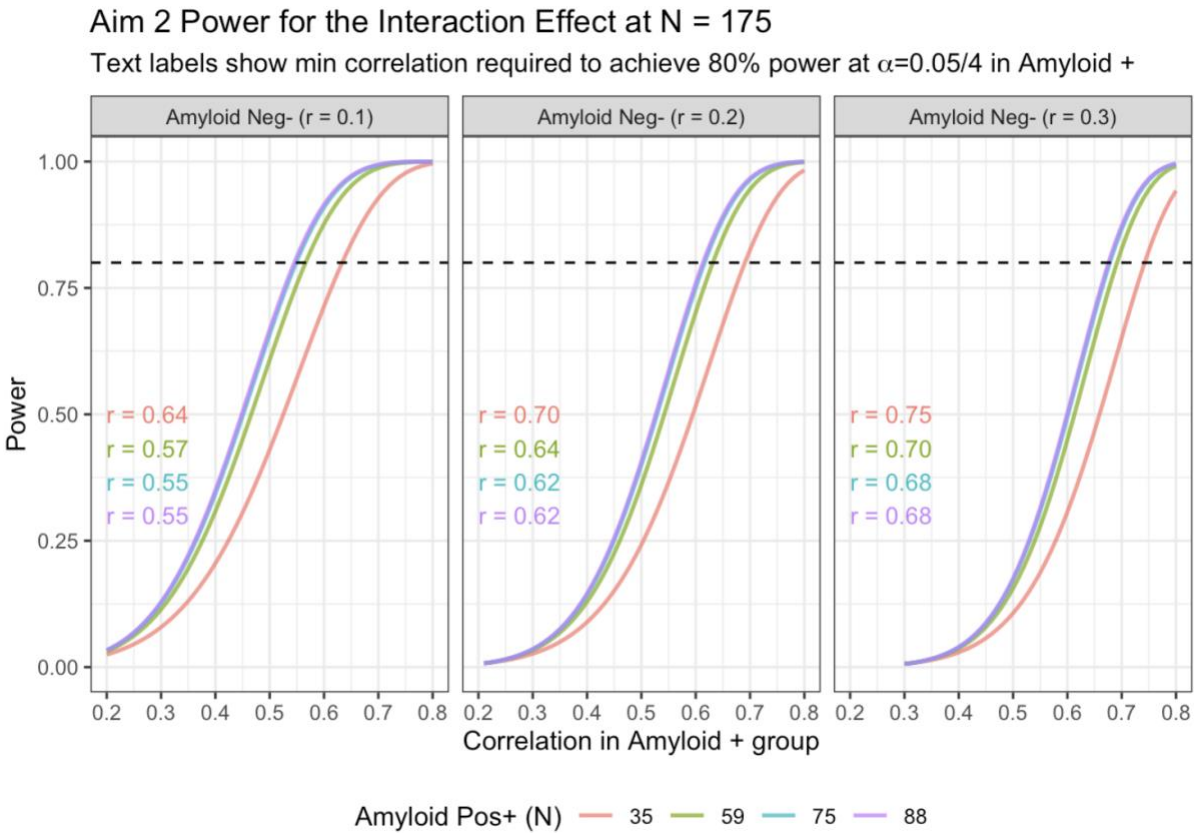

Figure *1*) indicated that power was driven primarily by differences in correlations between groups rather than differences in group sizes. Correlations in the amyloid-positive group of approximately r ≈ 0.55–0.70 were required to achieve 80% power, depending on the correlation and size differences between the amyloid groups.

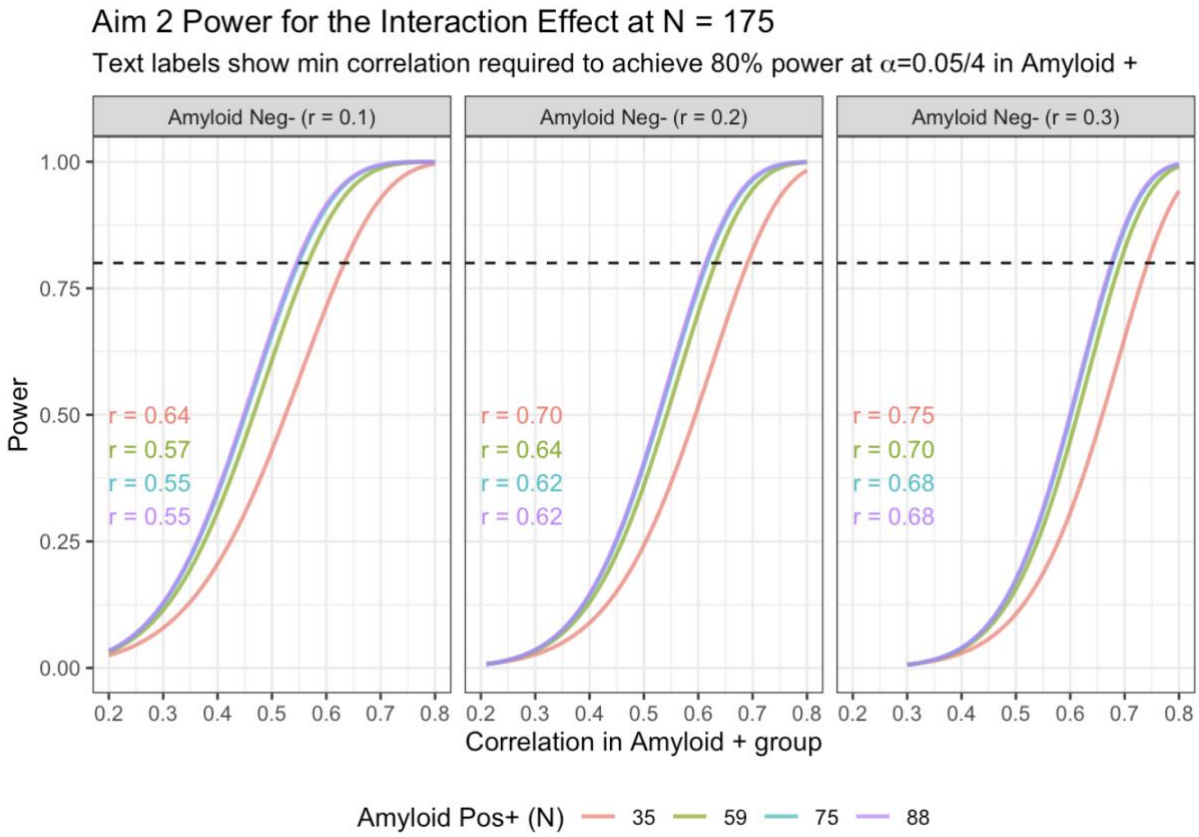

Figure 1. Aim 2 Power for Interaction Effects at N = 175. *Power curves illustrate the ability to detect differences in correlations between amyloid-negative and amyloid-positive groups at α = 0.0125. Panels correspond to increasing correlations in the amyloid-negative group (r = 0.1, 0.2, 0.3), while curves represent varying sample sizes in the amyloid-positive group.*

These results indicate that the proposed sample size of N=175 provides adequate power to detect moderate associations in Aim 1 and moderate-to-large interaction effects in Aim 2. Given that interaction effects are often smaller than main effects, the study will be underpowered to detect small interaction effects.